

SHARP REMAINDER ESTIMATES AND OPTIMAL LOCALIZATION FOR GURLAND'S RATIO OF THE GAMMA FUNCTION

ABSTRACT. Gurland's ratio $\mathcal{G}(x, y) = \Gamma(x)\Gamma(y)/\Gamma^2(\frac{x+y}{2})$ is the exponential of the Jensen gap of $\log \Gamma$ at the pair (x, y) . We show that this gap admits an exact power-series expansion in powers of $(\frac{x-y}{2})^2$ whose coefficients are Hurwitz zeta values — a classical identity, recorded in this form by Srivastava — and that the expansion converges *for every* $x, y > 0$: the natural radius of convergence is the distance from the midpoint $\frac{x+y}{2}$ to the pole of Γ at the origin, so the region of convergence is exactly the open quadrant. This answers in the affirmative two open problems recently posed by Wiśniewska (arXiv:2512.07028) and shows that the sufficient condition $Q < 1$ she imposed is redundant. We complement the expansion with two-sided remainder estimates whose geometric rate $\rho = |x - y|/(2 + x + y)$ we prove optimal; we show that the previously used rate Q satisfies $\rho \leq Q$ with equality exactly when $x = y$, and that Q is unbounded. We give the complete asymptotics of the ratio for large arguments in terms of the binary entropy function, extend the expansion to any number of variables, and derive applications to the beta function and to certified computation. Our main result localizes the mean-value parameter $t(x, y)$ of Wiśniewska; the two-sided endpoint bound that defines t appears already in the work of Yang and Zheng. We determine the asymptotics of t up to order $(x - y)^4$ and prove the sharp bound

$$\frac{t(x, y) - \sqrt{xy}}{\frac{x+y}{2} - \sqrt{xy}} > \frac{2}{3},$$

the constant $\frac{2}{3}$ being optimal and not attained. The proof is self-contained and uses neither of the two sufficient conditions (A) and (B) under which the weaker midpoint property $t > \frac{1}{2}(\sqrt{xy} + \frac{x+y}{2})$ had been established; we also record an independent, elementary route that yields the bound on an explicit range. We conjecture that conditions (A) and (B) cover the whole quadrant. All results are accompanied by high-precision numerical verification.

1. INTRODUCTION

1.1. **Background.** For $x, y > 0$ Gurland's ratio [5] is

$$(1) \quad \mathcal{G}(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma^2(\frac{x+y}{2})},$$

and Gurland's inequality states that $\mathcal{G}(x, y) > 1$ for $x \neq y$. Since $\log \Gamma$ is strictly convex on $(0, \infty)$ (Bohr–Mollerup), $\log \mathcal{G}(x, y)$ is exactly the *Jensen gap*

$$(2) \quad \log \mathcal{G}(x, y) = \log \Gamma(x) + \log \Gamma(y) - 2 \log \Gamma\left(\frac{x+y}{2}\right) \geq 0,$$

so Gurland's ratio is the exponential of the second-order Jensen gap of $\log \Gamma$. Following [9] we also consider the modified ratio

$$(3) \quad \mathcal{G}_\star(x, y) = \frac{\Gamma(1+x)\Gamma(1+y)}{\Gamma^2(1+\frac{x+y}{2})} = \frac{4xy}{(x+y)^2} \mathcal{G}(x, y).$$

The ratio (1) and its relatives were studied by Gautschi [4], Kairies [6], Merkle [8], Alzer [1] and Chen–Choi [2]; see [7] for a survey.

Writing

$$(4) \quad u := \frac{x-y}{2}, \quad v := \frac{x+y}{2}, \quad \text{so that } x = v+u, \quad y = v-u,$$

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Wiśniewska [9, Thm. 1], starting from the Weierstrass product, proved the finite expansion

$$(5) \quad \log \mathcal{G}_\star(x, y) = \sum_{k=1}^{m-1} \frac{1}{k} \left(\frac{x-y}{2} \right)^{2k} \zeta \left(2k, 1 + \frac{x+y}{2} \right) + R_n^m(x, y), \quad m \geq 2,$$

with the remainder estimate

$$(6) \quad 0 \leq R_n^m(x, y) \leq \varepsilon_n^m(x, y) = \frac{1}{m} \left(\frac{x-y}{2} \right)^{2m} \sum_{n=1}^{\infty} \frac{1}{(n+x)^m(n+y)^m},$$

where $\zeta(s, a) = \sum_{n \geq 0} (n+a)^{-s}$ is the Hurwitz zeta function. From (5)–(6) she deduced that the condition

$$(7) \quad Q(x, y) := \frac{|x-y|}{2(1+\sqrt{xy})} < 1$$

implies $\log \mathcal{G}_\star(x, y) = S_\infty(x, y)$, where $S_\infty = \lim_{m \rightarrow \infty} S_m$ and S_m denotes the partial sum in (5) [9, Cor. 1]; the bilateral inequality

$$(8) \quad \left(\frac{x-y}{2} \right)^2 \zeta \left(2, 1 + \frac{x+y}{2} \right) \leq \log \mathcal{G}_\star(x, y) \leq \left(\frac{x-y}{2} \right)^2 \zeta(2, 1 + \sqrt{xy})$$

[9, Cor. 2]. The two-sided bound (8) is, however, not new. Indeed, with the base point 1, $p = x$ and $q = y$, Corollary 11 and Remark 5 of Yang and Zheng [20] state that

$$(9) \quad \exp \left[\frac{(p-q)^2}{4} \psi'(1+r_1) \right] \leq \frac{\Gamma(1+p)\Gamma(1+q)}{\Gamma^2(1+\frac{p+q}{2})} \leq \exp \left[\frac{(p-q)^2}{4} \psi'(1+r_2) \right],$$

with the best constants $r_1 = \frac{p+q}{2}$ and $r_2 = \sqrt{pq}$ whenever $p, q > 0$; as $\psi' = \zeta(2, \cdot)$ and $\left(\frac{x-y}{2} \right)^2 = u^2$, this is exactly (8). Since $\zeta(2, \cdot)$ is continuous and strictly decreasing, (8) yields at once a unique parameter $t = t(x, y) \in (\sqrt{xy}, \frac{x+y}{2})$ with

$$(10) \quad \log \mathcal{G}_\star(x, y) = \left(\frac{x-y}{2} \right)^2 \zeta(2, 1+t) \quad [9, \text{Cor. 3}].$$

Thus Corollaries 2 and 3 of [9] follow from [20]. The question that remains is quantitative — where, inside $(\sqrt{xy}, \frac{x+y}{2})$, the point t lies — and it is this question that Theorems 1.8 and 1.9 answer. She then posed [9, §2] three problems.

Open problem 1. Does $R_n^m(x, y) \rightarrow 0$ hold for every $x, y > 0$?

Open problem 2. Does $\log \mathcal{G}_\star(x, y) = S_\infty(x, y)$ hold in general, and if not, how large can the difference be?

Open problem 3. How is $t(x, y)$ located in the interval $(\sqrt{xy}, \frac{x+y}{2})$? What is its distance to the midpoint of that interval?

1.2. Relation to the literature, and what is new here. The identity in Theorem 1.1 is a symmetrised Taylor expansion and is *classical*. Since $\psi^{(n)}(z) = (-1)^{n+1} n! \zeta(n+1, z)$ for $\text{Re } z > 0$ [3, §5.15], the Taylor series of $\log \Gamma$ about $a > 0$ is

$$\log \Gamma(a+w) = \log \Gamma(a) + \sum_{n \geq 1} \frac{(-1)^n}{n} \zeta(n, a) w^n, \quad |w| < a,$$

whose even part is exactly (12); equivalently,

$$(11) \quad \log \Gamma(a+w) + \log \Gamma(a-w) - 2 \log \Gamma(a) = \sum_{k \geq 1} \frac{\zeta(2k, a)}{k} w^{2k}, \quad |w| < a,$$

and it is in this form, as a sum of a series of Hurwitz zeta values, that the identity is recorded in the literature (see, e.g., [15]) and in monographs on zeta series [16]; the classical relation between the gamma function and the Hurwitz zeta function goes back at least to [11]. We therefore claim no novelty for (11), nor for its specialization $a = 1+v$, $w = u$, which is (13). What is new here is (i) the *sharp remainder theory* of the resulting convergent expansion: the two-sided estimate (15) with the *optimal* geometric rate ρ , together with the comparison with the previously used

rate Q ; (ii) the explicit observation that this specialization makes Wiśniewska's expansion (5) converge on the whole quadrant and her hypothesis $Q < 1$ redundant, which settles Open Problems 1 and 2 of [9]; (iii) the quantitative localization theory of the trigamma mean t : the bracket (19), the expansion (20) as $c \rightarrow 0$, and the sharp global bound $\theta > \frac{2}{3}$ of Theorem 1.9, whose best constant $\frac{2}{3}$ is not a corollary of the estimates of Theorem 1.8 but follows from a separate global argument in four steps; (iv) the large-argument asymptotics of Theorem 1.6; (v) the multivariate extension of Theorem 1.7; and (vi) the applications of Section 3.

Gurland's ratio has also been studied intensively by asymptotic methods. Merkle [8] derived bounds and asymptotic expansions; Alzer [10] and Qi [14] gave further bounds and a survey; Tian and Yang [17] established the asymptotic expansion

$$\frac{\Gamma(x+p)\Gamma(x+q)}{\Gamma(x+\frac{p+q}{2})^2} = \exp \left[\sum_{k=1}^n \frac{B_{2k}(s) - B_{2k}(\frac{1}{2})}{k(2k-1)(x+r_0)^{2k-1}} + R_n(x; p, q) \right], \quad x \rightarrow \infty,$$

where $p, q \in \mathbb{R}$, $w = |p - q| \neq 0$, $s = \frac{1-w}{2}$, $r_0 = \frac{p+q-1}{2}$ and B_{2k} are the Bernoulli polynomials, and proved that $(-1)^n R_n$ is completely monotone when $w < 1$, which yields sharp bounds; the remainder was analysed further by Yang and Tian [18] for the generalized ratio $\frac{\Gamma(x+a)\Gamma(x+b)}{\Gamma(x+c)\Gamma(x+d)}$ with $a+b = c+d$. These expansions are *asymptotic in $1/x$* with Bernoulli-polynomial coefficients, whereas (12)–(13) are *exact and convergent* in powers of $(\frac{x-y}{2})^2$ with Hurwitz-zeta coefficients. The two descriptions are complementary: the former is adapted to large arguments, the latter to arguments that are close together, and only the latter can be summed to the exact value. Refinements of Gurland's formula for π were given by Lin [12].

Two further strands of the literature are close to the localization problem studied here. Yang and Zheng [20] gave necessary and sufficient conditions for four classes of ratios of gamma functions to be logarithmically completely monotonic; their Corollary 11 and Remark 5 yield the two-sided bound (9), and their Propositions 3 and 4 compare double integrals of $\psi^{(n)}$ over rectangles, which is the same weighted-average picture that underlies (19). In [19] the mean $\psi_n^{-1}(\frac{1}{b-a} \int_a^b \psi_n(x+t) dt) - x$, generated by a *uniform* average of a polygamma function, is shown to increase from $\min(a, b)$ onto $\frac{a+b}{2}$; the parameter t of (10) is the corresponding mean for a *triangular* average of ψ' (see Remark 1.16), so the two objects are of the same type but not the same, and neither the existence of an interior sharp constant nor its value follows from [19]. Very recently, Moustafa and Mahmoud [13] obtained bounds for Gurland-type ratios in terms of the generalized Nielsen beta function. The sharp remainder theory of (12) and the quantitative localization theory developed below do not appear in these works.

1.3. Results of the present paper. Our starting point is that (2) is a second-order symmetric difference of $\log \Gamma$, so the subject is governed by the Taylor expansion of $\log \Gamma$ about the midpoint v . Since the singularities of $\log \Gamma$ nearest to $v > 0$ are the poles of Γ at $0, -1, -2, \dots$, the Taylor series about v converges on the disc $|z - v| < v$, and $x = v + u$, $y = v - u$ both lie in that disc precisely when $|u| < v$, i.e. precisely when $x, y > 0$. Thus the expansion converges on the whole quadrant and no additional hypothesis is required.

Theorem 1.1 (Exact expansion). *For all $x, y > 0$,*

$$(12) \quad \log \mathcal{G}(x, y) = \sum_{k=1}^{\infty} \frac{1}{k} \left(\frac{x-y}{2} \right)^{2k} \zeta \left(2k, \frac{x+y}{2} \right),$$

$$(13) \quad \log \mathcal{G}_\star(x, y) = \sum_{k=1}^{\infty} \frac{1}{k} \left(\frac{x-y}{2} \right)^{2k} \zeta \left(2k, 1 + \frac{x+y}{2} \right).$$

Both series converge absolutely. Moreover (12) converges for exactly those $(x, y) \in \mathbb{R}^2$ with $|x - y| < x + y$, and (13) for exactly those with $|x - y| < 2 + x + y$; on the quadrant $x, y > 0$ both hold identically, and the two regions are optimal.

Corollary 1.2. *For every $x, y > 0$ and every $m \geq 1$,*

$$\begin{aligned} R_n^m(x, y) &= \log \mathcal{G}_\star(x, y) - \sum_{k=1}^{m-1} \frac{1}{k} \left(\frac{x-y}{2} \right)^{2k} \zeta \left(2k, 1 + \frac{x+y}{2} \right) \\ &= \sum_{k=m}^{\infty} \frac{1}{k} \left(\frac{x-y}{2} \right)^{2k} \zeta \left(2k, 1 + \frac{x+y}{2} \right) \geq 0. \end{aligned}$$

Consequently $R_n^m(x, y) \downarrow 0$ and $\log \mathcal{G}_\star(x, y) = S_\infty(x, y)$ for every $x, y > 0$: Open Problems 1 and 2 both have affirmative answers. In particular the hypothesis $Q(x, y) < 1$ in [9, Cor. 1] is redundant.

The next result quantifies the convergence; the relevant rate is not (7) but

$$(14) \quad \rho(x, y) := \frac{|x-y|}{2+x+y} \quad (\mathcal{G}_\star), \quad \rho_{\mathcal{G}}(x, y) := \frac{|x-y|}{x+y} \quad (\mathcal{G}).$$

Theorem 1.3 (Sharp remainder estimates). *Let $x, y > 0$, $m \geq 1$, $u = \frac{x-y}{2}$, $v = \frac{x+y}{2}$, $\rho = \rho(x, y) = |x-y|/(2+x+y)$ and $R_m := \log \mathcal{G}_\star(x, y) - \sum_{k=1}^{m-1} \frac{u^{2k}}{k} \zeta(2k, 1+v)$. Then*

$$(15) \quad \frac{u^{2m}}{m} \zeta(2m, 1+v) \leq R_m \leq \frac{\rho^{2m}}{m(1-\rho^2)} \left(1 + \frac{1+v}{2m-1} \right).$$

Moreover, if $x \neq y$, then $\lim_{m \rightarrow \infty} \frac{1}{m} \log R_m = 2 \log \rho$, so the rate ρ in (15) is optimal. The same holds for $\mathcal{G}$ with ρ replaced by $\rho_{\mathcal{G}} = |x-y|/(x+y)$ and $\zeta(2m, 1+v)$ by $\zeta(2m, v)$.

Proposition 1.4. *For all $x, y > 0$ one has $\rho(x, y) \leq Q(x, y)$, with equality iff $x = y$. Moreover $\rho(x, y) < 1$ for all $x, y > 0$, whereas $\sup_{x, y > 0} Q(x, y) = +\infty$; for instance $Q(x, 1) = \frac{x-1}{2(1+\sqrt{x})} \sim \frac{\sqrt{x}}{2}$ as $x \rightarrow \infty$. Hence $\{Q < 1\}$ is a proper subset of the quadrant, and Corollary 1 of [9] does not cover the whole range of validity of the expansion.*

Corollary 1.5. *For all $x, y > 0$ and all $m \geq 1$,*

$$S_m(x, y) \leq \log \mathcal{G}_\star(x, y) \leq S_m(x, y) + \frac{\rho^{2m}}{m(1-\rho^2)} \left(1 + \frac{1+v}{2m-1} \right),$$

where $S_m(x, y) = \sum_{k=1}^{m-1} \frac{1}{k} \left(\frac{x-y}{2} \right)^{2k} \zeta(2k, 1+v)$. Taking $m = 2$ improves the upper bound of (8) whenever $\rho < \frac{\sqrt{5}}{3}$, while S_m increases to $\log \mathcal{G}_\star(x, y)$ as $m \rightarrow \infty$.

We also determine the behaviour of the ratio at large arguments. Recall

$$(16) \quad \Lambda(\rho) := (1+\rho) \log(1+\rho) + (1-\rho) \log(1-\rho), \quad |\rho| < 1.$$

One has $\Lambda(\rho) = \sum_{k \geq 1} \rho^{2k} / (k(2k-1))$, $\Lambda(\rho) > 0$ for $\rho \neq 0$, and $\Lambda(\rho) = 2\rho \operatorname{artanh} \rho + \log(1-\rho^2)$. As Λ vanishes to second order at the origin it is convenient to divide it by ρ^2 : we set

$$(17) \quad H(\rho) := \frac{\Lambda(\rho)}{\rho^2} = \sum_{k \geq 1} \frac{\rho^{2k-2}}{k(2k-1)} \quad (0 < \rho < 1), \quad H(0) := 1.$$

The series converges for $|\rho| < 1$, so H is continuous, strictly increasing and ≥ 1 on $[0, 1)$, and $\Lambda(\rho) = \rho^2 H(\rho)$ there.

Theorem 1.6 (Large-argument asymptotics). *Let $x_j, y_j > 0$, $v_j = \frac{x_j+y_j}{2} \rightarrow \infty$, $u_j = \frac{x_j-y_j}{2}$, and put $\rho_j := \frac{|u_j|}{1+v_j}$. Suppose*

$$\rho_j \longrightarrow \rho_0 \in [0, 1).$$

Then, with H as in (17),

$$\log \mathcal{G}_\star(x_j, y_j) = \frac{u_j^2}{1+v_j} H(\rho_j) (1 + o(1)) = (1+v_j) \Lambda(\rho_j) (1 + o(1)) \quad (j \rightarrow \infty),$$

and the analogous statement holds for $\log \mathcal{G}$ with $1 + v_j$ replaced by v_j throughout, i.e. under the hypothesis $|u_j|/v_j \rightarrow \rho_0 \in [0, 1)$ one has $\log \mathcal{G} = \frac{u_j^2}{v_j} H\left(\frac{|u_j|}{v_j}\right)(1 + o(1))$. (The sign of u_j is immaterial, since Λ and H are even.)

The expansion extends verbatim to any number of variables.

Theorem 1.7 (Multivariate expansion). *Let $n \geq 2$, $x_1, \dots, x_n > 0$, $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ and $m_j = \sum_{i=1}^n (x_i - \bar{x})^j$ for $j \geq 2$. If $\max_i |x_i - \bar{x}| < \bar{x}$, then*

$$(18) \quad \log \frac{\prod_{i=1}^n \Gamma(x_i)}{\Gamma(\bar{x})^n} = \sum_{j=2}^{\infty} \frac{(-1)^j}{j} \zeta(j, \bar{x}) m_j,$$

the series converging absolutely. In particular $\prod_{i=1}^n \Gamma(x_i) \geq \Gamma(\bar{x})^n$, with equality iff $x_1 = \dots = x_n$.

Finally we address Open Problem 3. We first note that t is uniquely determined, then we control its distance to the midpoint by two independent arguments, and we determine the sharp constant in the resulting bound.

Theorem 1.8 (Localization of t). *Let $x \neq y$ be positive, $v = \frac{x+y}{2}$, $s = \sqrt{xy}$, $c = \frac{|x-y|}{2}$, $A = 1 + v$ and $\rho = c/A$. Then:*

- (i) *there is a unique $t = t(x, y) \in (s, v)$ satisfying (10);*
- (ii) *with $\rho = \rho(x, y)$ one has*

$$(19) \quad \frac{c^2 \zeta(4, A)}{4\zeta(3, A) + 6v\zeta(4, A)} \leq v - t \leq \frac{c^2 \zeta(4, A)}{4\zeta(3, A)} \left(1 + \frac{2\rho^2}{3(1 - \rho^2)}\right);$$

- (iii) *as $c \rightarrow 0^+$ with v fixed,*

$$(20) \quad v - t = \frac{c^2 \zeta(4, A)}{4\zeta(3, A)} + \frac{c^4}{2\zeta(3, A)} \left[\frac{\zeta(6, A)}{3} - \frac{3\zeta(4, A)^3}{16\zeta(3, A)^2} \right] + O(c^6);$$

- (iv) *$t > \frac{1}{2}(\sqrt{xy} + \frac{x+y}{2})$ whenever at least one of the following two conditions holds:*

$$(21) \quad (A) \quad \frac{v + s}{2(1 + v)} \left(1 + \frac{2\rho^2}{3(1 - \rho^2)}\right) < 1,$$

$$(22) \quad (B) \quad \mathcal{E}(x, y) := \int_{v+\frac{1}{2}}^{\infty} K - \frac{1}{A - w} - \frac{1}{2(A - w)^2} < 0,$$

where $w = \frac{v-s}{2}$ and $K(\xi) = \frac{1}{c^2} \log \frac{1}{1 - c^2/\xi^2}$ for $\xi > c$.

Condition (A) is a Hurwitz-zeta comparison; condition (B) is a convexity bound, and $\int_B^{\infty} K$ has the closed form

$$(23) \quad \int_B^{\infty} K = \frac{1}{c^2} \left[-B \log \frac{B^2}{B^2 - c^2} + c \log \frac{B + c}{B - c} \right], \quad B > c.$$

The two conditions are complementary — (A) is effective for moderate ρ , (B) as $\rho \rightarrow 1$ — and numerically their union covers the whole parameter range (Section 4). This leads us to sharpen the second half of Open Problem 3.

Theorem 1.9. *For all $x \neq y > 0$,*

$$(24) \quad \frac{t(x, y) - \sqrt{xy}}{\frac{x+y}{2} - \sqrt{xy}} > \frac{2}{3},$$

equivalently $t(x, y) > \frac{1}{3}(x + y + \sqrt{xy})$. The constant $\frac{2}{3}$ is best possible and the bound is not attained: the ratio in (24) tends to $\frac{2}{3}$ along $v \rightarrow \infty$, $|x - y|/(x + y) \rightarrow 0$.

Proof. Fix $v > 0$, let $0 < c < v$, $s = \sqrt{v^2 - c^2}$, $A = 1 + v$, $\Delta = \frac{v-s}{3}$ and $\Phi(c) = \log \mathcal{G}_*(x, y)/c^2 = \sum_{k \geq 1} \frac{c^{2k-2}}{k} \zeta(2k, A)$. Since $\zeta(2, 1+t) = \Phi(c)$ and $\zeta(2, \cdot)$ is strictly decreasing,

$$\theta > \frac{2}{3} \iff t > \frac{2v+s}{3} = v - \Delta \iff \zeta(2, A - \Delta) > \Phi(c) \iff R(c) > 0,$$

and $R(0) = 0$ by continuity. We prove $R'(c) > 0$ on $(0, v)$. Because $\Delta' = \frac{c}{3s}$ and $\partial_\Delta \zeta(2, A - \Delta) = 2\zeta(3, A - \Delta)$,

$$(25) \quad R'(c) = \frac{2c}{3s} \zeta(3, A - \Delta) - \frac{2}{c} E(c), \quad E(c) = \sum_{m \geq 1} \frac{m}{m+1} c^{2m} \zeta(2m+2, A),$$

so it suffices to prove

$$(26) \quad E(c) < \frac{c^2}{3s} \zeta(3, A - \Delta).$$

Step 1 (a global midpoint bound for E). Put $K_c(u) = \frac{1}{c^2} \log \frac{1}{1-c^2/u^2}$ and $Q_c(u) = \frac{1}{u^2-c^2} - K_c(u)$. From $K_c(u) = \sum_{j \geq 0} \frac{c^{2j}}{j+1} u^{-2j-2}$ and $\frac{1}{u^2-c^2} = \sum_{j \geq 0} c^{2j} u^{-2j-2}$,

$$Q_c(u) = \sum_{m \geq 1} \frac{m}{m+1} c^{2m} u^{-2m-2} > 0, \quad E(c) = \sum_{n \geq 0} Q_c(A+n).$$

All coefficients being positive, Q_c is positive, decreasing and convex on $u > c$; the midpoint rule for a convex function, summed over $n \geq 0$, gives with $B = A - \frac{1}{2} > c$

$$(27) \quad E(c) < \int_B^\infty Q_c(u) du,$$

with no restriction on the size of c^2/A . Term-by-term integration with $y = c^2/B^2 \in (0, 1)$ gives

$$(28) \quad \int_B^\infty Q_c(u) du = \frac{c^2}{6B^3} F(y), \quad F(y) = 6 \sum_{m \geq 1} \frac{m}{(m+1)(2m+1)} y^{m-1}.$$

Step 2 (a sharp one-variable inequality). Let $S(y) = \sum_{m \geq 0} \frac{y^m}{(m+1)(2m+1)}$, so $F = 6S'$, and let $h(y) = \frac{1-\sqrt{1-y}}{2+\sqrt{1-y}}$, for which a direct differentiation gives $6h'(y) = \frac{9}{\sqrt{1-y}(2+\sqrt{1-y})^2}$. The identity $h = \frac{3(1-\sqrt{1-y})-y}{3+y}$, together with $1 - \sqrt{1-y} = \sum_{m \geq 1} c_m y^m$, $c_m = \frac{\binom{2m}{m}}{4^m(2m-1)}$, $\frac{c_m}{c_{m-1}} = \frac{2m-3}{2m}$, shows that $h = \sum_{m \geq 1} a_m y^m$ with $a_1 = \frac{1}{6}$ and $a_m = c_m - \frac{1}{3}a_{m-1}$ for $m \geq 2$. Writing $S = 1 + \sum_{m \geq 1} b_m y^m$, $b_m = \frac{1}{(m+1)(2m+1)}$, we have $a_1 = b_1 = \frac{1}{6}$ and, for $r_m = a_m/c_m$, $r_m = 1 - \frac{2m}{3(2m-3)}r_{m-1}$ with $r_2 = \frac{5}{9}$, $r_3 = \frac{17}{27}$; a two-line induction gives $\frac{3(m-1)}{4m} \leq r_m < \frac{3}{4}$ for every $m \geq 2$. Hence $a_2 - b_2 = \frac{1}{360}$, $a_3 - b_3 = \frac{11}{3024}$, and for $m \geq 4$ $a_m > c_m - \frac{1}{4}c_{m-1} = \frac{3(m-2)}{2(2m-3)}c_m$, while $Q_m := \frac{3(m-2)c_m}{2(2m-3)b_m}$ satisfies $Q_4 = \frac{135}{128} > 1$ and $\frac{Q_{m+1}}{Q_m} = \frac{(m-1)(m+2)(2m-3)(2m+3)}{2(m-2)(m+1)^2(2m+1)} > 1$ because $2m^3 - 5m^2 + 5m + 22 > 0$. Thus $a_m > b_m$ for all $m \geq 2$, so $h' > S'$ and

$$(29) \quad F(y) < \frac{9}{\sqrt{1-y}(2+\sqrt{1-y})^2}, \quad 0 < y < 1.$$

Step 3 (the finite- v correction only helps). Let $z = A - \Delta$, $e = \frac{1}{2B} \in (0, 1)$ and $q = \frac{s}{B}$, so that $q = \sqrt{(1-e)^2 - y}$ and, exactly, $z = \frac{B(2+q+4e)}{3}$. A direct computation gives

$$\frac{B^3(z+1)}{sz^3} = L_e(y) := \frac{9(2+q+10e)}{q(2+q+4e)^3},$$

and $L_0(y)$ is precisely the right-hand side of (29). Since $q'(e) = -\frac{1-e}{q}$,

$$\frac{d}{de} \log L_e = -\frac{N(e, q)}{q^2(2+q+4e)(2+q+10e)},$$

where $N(e, q) = 40e^3 + 40e^2q - 12e^2 + 83eq^2 - 32eq - 24e + 2q^3 + q^2 - 8q - 4$. For fixed $e \in (0, 1)$ the function $q \mapsto N(e, q)$ is strictly convex ($\partial_q^2 N = 166e + 12q + 2 > 0$), and on

$0 < q < 1 - e$ its endpoint values are $N(e, 0) = 4(e - 1)(2e + 1)(5e + 1) < 0$ and $N(e, 1 - e) = 9(e - 1)(9e^2 - 2e + 1) < 0$. A convex function lies below the maximum of its endpoint values, so $N(e, q) < 0$ on $0 < q < 1 - e$; hence $e \mapsto L_e(y)$ is strictly increasing and

$$(30) \quad \frac{9}{\sqrt{1-y}(2+\sqrt{1-y})^2} < L_e(y) = \frac{B^3(z+1)}{sz^3}.$$

Step 4 (conclusion). Combining (27)–(30),

$$E(c) < \frac{c^2}{6B^3} F(y) < \frac{c^2}{6B^3} \cdot \frac{B^3(z+1)}{sz^3} = \frac{c^2(z+1)}{6sz^3}.$$

Finally, the trapezoid rule applied to the convex decreasing function $t \mapsto (z+t)^{-3}$ gives $\zeta(3, z) \geq \int_0^\infty (z+t)^{-3} dt + \frac{1}{2z^3} = \frac{z+1}{2z^3}$, so $E(c) < \frac{c^2}{3s} \zeta(3, z)$, which is (26). Therefore $R'(c) > 0$ on $(0, v)$; with $R(0) = 0$ this gives $R(c) > 0$, and by (25) and the first display $\theta > \frac{2}{3}$, as claimed. $\square$

Conjecture 1.10. Conditions (A) and (B) of Theorem 1.8 are satisfied for every $x \neq y > 0$.

Theorem 1.11 (An explicit certified range). *Let $x \neq y > 0$ satisfy*

$$(31) \quad (x-y)^2 \leq \frac{2+x+y}{3},$$

equivalently $c^2 \leq \frac{A}{6}$, where $c = \frac{|x-y|}{2}$ and $A = 1+v$. Then $\theta(x, y) > \frac{2}{3}$. In particular, for every fixed $d \neq 0$ inequality (24) holds for all $x, y > 0$ with $|x-y| = d$ and $x+y$ sufficiently large: this is precisely the range in which the constant $\frac{2}{3}$ is sharp.

Remark 1.12 (A product formula). The Weierstrass product for Γ gives a representation of $\mathcal{G}_\star$ that makes the identity $\Phi = \sum_{m \geq 0} K(A+m)$ transparent. Indeed $\Gamma(z) = \frac{e^{-\gamma z}}{z} \prod_{n \geq 1} \frac{e^{z/n}}{1+z/n}$ yields, with $v = \frac{x+y}{2}$ and $c = \frac{|x-y|}{2}$,

$$\frac{\Gamma(v+c)\Gamma(v-c)}{\Gamma(v)^2} = \frac{v^2}{v^2-c^2} \prod_{n \geq 1} \frac{(n+v)^2}{(n+v)^2-c^2},$$

the factors $e^{\pm z/n}$ cancelling. Since $\mathcal{G}_\star(x, y) = \frac{4xy}{(x+y)^2} \mathcal{G}(x, y) = \frac{v^2-c^2}{v^2} \cdot \frac{\Gamma(v+c)\Gamma(v-c)}{\Gamma(v)^2}$, the factors $v^2 - c^2$ cancel as well and

$$(32) \quad \mathcal{G}_\star(x, y) = \prod_{n \geq 1} \frac{(n+v)^2}{(n+v)^2-c^2} = \prod_{m \geq 0} \frac{(A+m)^2}{(A+m)^2-c^2},$$

Taking logarithms, and writing $A = 1+v$, recovers $\log \mathcal{G}_\star = \sum_{m \geq 0} \log(1 - \frac{c^2}{(A+m)^2})^{-1} = c^2 \sum_{m \geq 0} K(A+m) = \sum_{k \geq 1} \frac{c^{2k}}{k} \zeta(2k, A)$, i.e. Theorem 1.1 for $\mathcal{G}_\star$; each factor of (32) is positive because $c < v < A$. In particular $\mathcal{G}_\star$ is a product of terms > 1 , which is the Jensen–Gurland inequality in multiplicative form. The individual summand in $R(c)$ is positive once $u > \frac{c^2+\Delta^2}{2\Delta}$: from $-\log(1-y) \leq \frac{y}{1-y}$ with $y = \frac{c^2}{u^2}$ it is enough that $u^2 - c^2 > (u - \Delta)^2$, i.e. $u > \frac{c^2+\Delta^2}{2\Delta}$. So only the $\max\{0, \lfloor \frac{c^2+\Delta^2}{2\Delta} - A \rfloor + 1\}$ smallest indices can be exceptional, and numerically their total deficit is exceeded by the total surplus of the others by a factor that falls to 1.0016 at $v = 10^4$, $c = 0.1v$. This is the quantitative reason why estimates coarser than about 10^{-3} relative accuracy cannot decide Theorem 1.9.

Remark 1.13 (A monotone formulation). Theorem 1.9 would also follow from the combination of the following two statements about the single quantity $\Theta(c) := \frac{v-t(c)}{v-s(c)}$, $0 < c < v$, where

$t(c)$ is the point with $\zeta(2, 1+t) = \log \mathcal{G}_\star/c^2$ and $s = \sqrt{v^2 - c^2}$:

- (m1) Θ is strictly decreasing on $(0, v)$;
- (m2) $\lim_{c \rightarrow 0^+} \Theta(c) \leq \frac{1}{3}$.

Indeed (m1) and (m2) give $\Theta(c) < \frac{1}{3}$ for every $c > 0$, which is $v - t < \frac{v-s}{3}$, i.e. $\theta > \frac{2}{3}$. Statement (m2) is a local computation: as $c \rightarrow 0$ one has $v - t \sim \frac{c^2 \zeta(4, A)}{4\zeta(3, A)}$ by (20) and $v - s \sim \frac{c^2}{2v}$, so

$$\lim_{c \rightarrow 0^+} \Theta(c) = \frac{v \zeta(4, A)}{2 \zeta(3, A)} = \frac{1}{3} \left(1 - \frac{1}{2A} + O(A^{-2}) \right) \quad (A \rightarrow \infty),$$

and this limit is $< \frac{1}{3}$ for every $A > 1$: by (38), $\frac{\zeta(4, A)}{\zeta(3, A)} \leq \frac{2A^3}{3(A - \frac{1}{2})^3(A+1)}$, so $\lim_{c \rightarrow 0^+} \Theta(c) < \frac{1}{3}$ is equivalent to $(A - 1)A^3 < (A - \frac{1}{2})^3(A + 1)$, i.e. to $4A^3 - 6A^2 + 5A - 1 > 0$, which holds for $A \geq 1$. Statement (m1) is what remains: numerically it holds for every tested v (1.5, 2, 5, 20, 100, 10^3 , 10^4), Θ decreasing from $\frac{1}{3} - \frac{1}{6A}$ at $c \rightarrow 0$ to values as low as 0.183 at $c \rightarrow v$ for $v = \frac{3}{2}$. Equivalently, by $\zeta(2, A - \delta) = \Phi$ and $\delta' = E/(c\zeta(3, A - \delta)) > 0$, while $\Delta' = c/(3s)$, so (m1) reads

$$(33) \quad \delta(c) c^2 \zeta(3, A - \delta(c)) > s(c) (v - s(c)) E(c), \quad E(c) = \sum_{m \geq 1} \frac{m}{m+1} c^{2m} \zeta(2m+2, A).$$

Remark 1.14 (A wider certified range). The proof of Theorem 1.11 can be repeated with a sharper split of the logarithm. Since $-\log(1 - y) = y + \frac{y^2}{2} + \sum_{k \geq 3} \frac{y^k}{k} \leq y + \frac{y^2}{2} + \frac{y^3}{3(1-y)}$ for $0 < y < 1$, and since $(u - \Delta)^2 \leq u^2$, one has

$$c^2 R(c) \geq c^2 [2\Delta \zeta(3, A) - \Delta^2 \zeta(4, A)] - \frac{c^4 \zeta(4, A)}{2(1 - \rho^2)},$$

so $R(c) > 0$ certainly holds as soon as

$$(34) \quad 2\Delta \zeta(3, A) > \Delta^2 \zeta(4, A) + \frac{c^2 \zeta(4, A)}{2(1 - \rho^2)}, \quad \Delta = \frac{v - s}{3}, \quad \rho = \frac{c}{A}.$$

Condition (34) is strictly weaker than the hypothesis of Theorem 1.11: it holds throughout $c^2 \lesssim 0.62A$ (numerically $c^2/A = 0.8145, 0.6661, 0.6297, 0.6216, 0.6208$ for $v = 5, 20, 100, 1000, 10^4$), whereas $c^2 \leq \frac{A}{6} \approx 0.167A$ was all that the earlier estimates gave. Beyond $c^2 \approx 0.62A$ the right-hand side of (34) overtakes the left and this particular chain stops discriminating.

Remark 1.15 (Scaling limit). Fix $\rho = \frac{c}{A}$ and let $A \rightarrow \infty$. Since $\zeta(2m+2, A) \sim \frac{A^{-(2m+1)}}{2m+1}$ and $v - s = A(1 - \sqrt{1 - \rho^2})(1 + o(1))$, the comparison $\zeta(2, A - \Delta) > \Phi$ behind Theorem 1.9 becomes, at leading order in A^{-1} ,

$$(35) \quad \frac{t}{1-t} > \Sigma_0(\rho) - 1, \quad t = \frac{1 - \sqrt{1 - \rho^2}}{3}, \quad \Sigma_0(\rho) = \sum_{m \geq 0} \frac{\rho^{2m}}{(m+1)(2m+1)},$$

and Σ_0 has the closed form $\Sigma_0(\rho) = 2 \frac{\operatorname{artanh} \rho}{\rho} + \frac{\log(1 - \rho^2)}{\rho^2}$. The difference of the two sides of (35), $D(\rho) = \frac{t}{1-t} - \Sigma_0(\rho) + 1$, is positive on $(0, 1)$, and this is again a consequence of Step 2 of the proof of Theorem 1.9. Indeed, with $y = \rho^2$,

$$\frac{t}{1-t} = \frac{1 - \sqrt{1-y}}{2 + \sqrt{1-y}} = h(y), \quad \Sigma_0(\rho) = S(y), \quad \text{so } D = h - S + 1.$$

Step 2 writes $h = \sum_{m \geq 1} a_m y^m$ and $S = 1 + \sum_{m \geq 1} b_m y^m$ with $a_1 = b_1 = \frac{1}{6}$ and $a_m > b_m$ for every $m \geq 2$; hence $D = \sum_{m \geq 2} (a_m - b_m) y^m$ has positive coefficients, which proves $D > 0$ and at the same time identifies the expansion $D(\rho) = \frac{\rho^4}{360} + \frac{11\rho^6}{3024} + O(\rho^8)$, since $a_2 - b_2 = \frac{1}{360}$ and $a_3 - b_3 = \frac{11}{3024}$ are the coefficients computed there (with a positive $O(\rho^8)$ tail). Numerically $D(\rho) \approx \rho^4/360$ down to $\rho = 5 \cdot 10^{-4}$ (where $D = 1.74 \cdot 10^{-16}$) and $D(0.99) = 5.10 \cdot 10^{-2}$. Thus the constant $\frac{2}{3}$ is decided by an $O(\rho^4)$ cancellation: this is why no first-order estimate settles Theorem 1.9.

Remark 1.16. Theorem 1.9 can be restated as a comparison of $\zeta(2, \cdot)$ with its own triangular average. With $A = 1 + v$, $c = |x - y|/2$, K as in (23) and $\Phi = \log \mathcal{G}_\star(x, y)/c^2$, the identity

$\sum_{n \geq 0} K(A+n) = \Phi$ of the proof of Theorem 1.8(iv) combines with $\psi(A+t) - \psi(A-t) = \int_{-t}^t \psi'(A+z) dz$ to give

$$\Phi = \frac{1}{c^2} \int_0^c [\psi(A+t) - \psi(A-t)] dt = \int_{-c}^c \psi'(A+z) \frac{c-|z|}{c^2} dz.$$

Since $\zeta(2, 1+t) = \psi'(1+t)$, the inequality of Theorem 1.9 is therefore equivalent to

$$\int_{-c}^c \left[\psi'(A-\Delta) - \psi'(A+z) \right] \frac{c-|z|}{c^2} dz > 0, \quad \Delta = \frac{v-s}{3} = \frac{c^2}{3(v+s)},$$

where $s = \sqrt{xy}$: it compares the value of $\psi' = \zeta(2, \cdot)$ at the point $A - \Delta$ with its average over $[-c, c]$ against the triangular weight $(c-|z|)/c^2$, whose variance is $c^2/6$. Expanding both sides about A , the terms of order c^0 and c^1 cancel (the weight is even), and the first surviving comparison is

$$\frac{\zeta(4, A)}{\zeta(3, A)} < \frac{4}{3(v+s)},$$

which holds for all $x, y > 0$, with relative margin $\frac{1}{A}$ because $\zeta(4, A)/\zeta(3, A) = \frac{2}{3A}(1 + O(A^{-1}))$ and $s \leq v$. The inequality is delicate because the decision rests on the next terms, of relative size ρ^2 : this is also why the constant $\frac{2}{3}$ is sharp but not attained.

Remark 1.17. The limiting statement in Theorem 1.9 follows from Theorem 1.8(iii). Take $x_j = v_j(1 + \varepsilon_j)$, $y_j = v_j(1 - \varepsilon_j)$ with $v_j \rightarrow \infty$, $\varepsilon_j \rightarrow 0$ and $v_j \varepsilon_j \rightarrow 0$. Then $c_j = v_j \varepsilon_j \rightarrow 0$, so (20) applies and, since $\zeta(4, A)/\zeta(3, A) \sim 2/(3A)$ as $A \rightarrow \infty$,

$$v_j - t_j \sim \frac{v_j^2 \varepsilon_j^2}{4} \cdot \frac{2}{3(1 + v_j)} \sim \frac{v_j \varepsilon_j^2}{6}, \quad v_j - s_j = \frac{v_j \varepsilon_j^2}{1 + \sqrt{1 - \varepsilon_j^2}} \sim \frac{v_j \varepsilon_j^2}{2}.$$

Hence $(v_j - t_j)/(v_j - s_j) \rightarrow \frac{1}{3}$ and $(t_j - s_j)/(v_j - s_j) \rightarrow \frac{2}{3}$. Thus $\theta = \frac{2}{3} + \frac{1}{6v} + O(v^{-2})$; the penultimate column of Table 5 is such a path and confirms this to nine significant digits. It also shows that a fixed ε does *not* suffice: there $\theta \rightarrow \frac{2}{3} + g(\varepsilon) > \frac{2}{3}$.

2. PROOFS

2.1. Proof of Theorem 1.1. Let $v > 0$ and put $\Omega = \{z \in \mathbb{C} : \operatorname{Re} z > 0\}$. The principal branch of $\log \Gamma$ is holomorphic on Ω , and the disc $D(v, v) = \{z : |z - v| < v\}$ is contained in Ω . The nearest singularities of $\log \Gamma$ to v are the poles of Γ at $0, -1, -2, \dots$, all at distance $\geq v$ from v , and $|0 - v| = v$; hence the Taylor series of $\log \Gamma$ about v has radius of convergence exactly v and converges to $\log \Gamma$ on $D(v, v)$.

Fix u with $|u| < v$. Then $v \pm u \in D(v, v)$ and

$$\log \Gamma(v+w) = \log \Gamma(v) + \sum_{n=1}^{\infty} \frac{\psi^{(n-1)}(v)}{n!} w^n, \quad |w| < v,$$

where $\psi = \Gamma'/\Gamma$ and $\psi^{(n)}$ are the polygamma functions. Evaluating at $w = u$ and $w = -u$ and adding,

$$\log \Gamma(v+u) + \log \Gamma(v-u) - 2 \log \Gamma(v) = 2 \sum_{k=1}^{\infty} \frac{\psi^{(2k-1)}(v)}{(2k)!} u^{2k}.$$

The relation $\psi^{(n)}(z) = (-1)^{n+1} n! \zeta(n+1, z)$ for $\operatorname{Re} z > 0$ (see e.g. [3, §5.15]) gives, for $n = 2k-1$, $\psi^{(2k-1)}(v) = (2k-1)! \zeta(2k, v)$. Therefore

$$\log \mathcal{G}(x, y) = 2 \sum_{k=1}^{\infty} \frac{(2k-1)!}{(2k)!} \zeta(2k, v) u^{2k} = \sum_{k=1}^{\infty} \frac{u^{2k}}{k} \zeta(2k, v),$$

which is (12); its region of convergence is $|u| < v$, i.e. $\{|x-y| < x+y\}$, and since all terms are non-negative the convergence is absolute. The same argument applied to $z \mapsto \log \Gamma(1+z)$ at the point $1+v$, whose singularities are $\{-1, -2, -3, \dots\}$ and hence lie at distance $\geq 1+v$

from $1 + v$, with equality at -1 , yields (13) with convergence exactly for $|u| < 1 + v$, i.e. for $|x - y| < 2 + x + y$. Finally, $|x - y| < x + y$ and $|x - y| < 2 + x + y$ both hold for all $x, y > 0$; the first fails for $(x, y) = (1, -\frac{1}{2})$ and the second for $(x, y) = (3, -2)$, so the stated regions are optimal. $\square$

2.2. Proof of Corollary 1.2. By Theorem 1.1, for every $x, y > 0$ the series (13) converges and all its terms are positive. Hence $R_n^m(x, y) = \sum_{k=m}^{\infty} \frac{u^{2k}}{k} \zeta(2k, 1+v)$ is the tail of a convergent series of positive terms; it is positive and decreases to 0. This is precisely Open Problems 1 and 2. The final claim is Proposition 1.4. $\square$

2.3. Proof of Theorem 1.3. Put $A = 1 + v$, $c = |u|$, $\rho = c/A \in (0, 1)$. We use the integral-test estimate

$$(36) \quad \zeta(2k, A) \leq A^{-2k} \left(1 + \frac{A}{2k-1}\right), \quad k \geq 1, A > 0,$$

obtained from $\zeta(2k, A) = \sum_{n \geq 0} (n + A)^{-2k} \leq A^{-2k} + \int_0^{\infty} (t + A)^{-2k} dt$.

Lower bound. All terms of the tail are positive, so $R_m \geq \frac{u^{2m}}{m} \zeta(2m, A)$.

Upper bound. By (36),

$$R_m \leq \sum_{k=m}^{\infty} \frac{c^{2k}}{k} A^{-2k} \left(1 + \frac{A}{2k-1}\right) \leq \left(1 + \frac{A}{2m-1}\right) \sum_{k=m}^{\infty} \frac{\rho^{2k}}{k} = \left(1 + \frac{A}{2m-1}\right) \rho^{2m} \sum_{j=0}^{\infty} \frac{\rho^{2j}}{m+j},$$

and $\sum_{j \geq 0} \frac{\rho^{2j}}{m+j} \leq \frac{1}{m(1-\rho^2)}$, which is (15) for $\mathcal{G}_\star$.

Optimality. Let $x \neq y$, so $c > 0$ and $\rho \in (0, 1)$. Write $\zeta(2m, A) = A^{-2m}(1 + \varepsilon_m)$; since $\sum_{n \geq 1} \left(\frac{A}{n+A}\right)^{2m} \leq \int_0^{\infty} \left(\frac{A}{A+t}\right)^{2m} dt = \frac{A}{2m-1}$, we have $0 \leq \varepsilon_m \leq \frac{A}{2m-1} \rightarrow 0$. Hence $R_m \geq \frac{c^{2m}}{m} \zeta(2m, A) = \frac{1+\varepsilon_m}{m} \rho^{2m}$, so $\frac{1}{m} \log R_m \geq 2 \log \rho + o(1)$; conversely (15) gives $\frac{1}{m} \log R_m \leq 2 \log \rho + \frac{1}{m} \log \frac{1+\frac{A}{2m-1}}{m(1-\rho^2)} = 2 \log \rho + o(1)$. The proof for $\mathcal{G}$ is identical with A replaced by v ; there $\zeta(2k, v) \leq v^{-2k} \left(1 + \frac{v}{2k-1}\right)$ and $\rho_{\mathcal{G}} = c/v$. $\square$

2.4. Proof of Proposition 1.4. Since $x + y \geq 2\sqrt{xy}$, $\rho = \frac{|x-y|}{2+x+y} \leq \frac{|x-y|}{2+2\sqrt{xy}} = Q$, with equality iff $x = y$. Also $|x - y| < x + y < 2 + x + y$, so $\rho < 1$. Finally $Q(x, 1) = \frac{x-1}{2(1+\sqrt{x})} \sim \frac{\sqrt{x}}{2} \rightarrow \infty$. $\square$

2.5. Proof of Corollary 1.5. The two-sided inequality is Theorem 1.3 restated, and $S_m \rightarrow \log \mathcal{G}_\star(x, y)$ as $m \rightarrow \infty$ by Corollary 1.2. It remains to compare, at $m = 2$, the correction in (15) with the gain over the upper bound of (8). Write $s = \sqrt{xy}$, $c = |u|$ and $A = 1 + v$, so that $\rho = c/A$ and

$$c^2 [\zeta(2, 1+s) - \zeta(2, 1+v)] = 2c^2 \int_{1+s}^{1+v} \zeta(3, z) dz \geq 2c^2 \int_{1+s}^{1+v} \frac{dz}{2z^2} = \frac{c^2(v-s)}{(1+s)(1+v)},$$

because $\zeta(3, z) = \sum_{n \geq 0} (n+z)^{-3} \geq \int_0^{\infty} (t+z)^{-3} dt = \frac{1}{2z^2}$ (the integrand decreases, so $\int_n^{n+1} (t+z)^{-3} dt \leq (n+z)^{-3}$ for every $n \geq 0$). Since $v-s = \frac{v^2-s^2}{v+s} = \frac{c^2}{v+s}$ and $(v+s)(1+s) \leq 2v(1+v)$, the gain is at least $\frac{c^4}{2v(1+v)^2}$. The correction in (15) at $m = 2$ is

$$\frac{\rho^4}{2(1-\rho^2)} \left(1 + \frac{A}{3}\right) = \frac{c^4(A+3)}{6A^4(1-\rho^2)},$$

which is smaller as soon as $\frac{v(A+3)}{3A^2(1-\rho^2)} < 1$. As $\frac{v(A+3)}{3A^2} = \frac{v(v+4)}{3(v+1)^2} \leq \frac{4}{9}$, with equality at $v = 2$, this holds whenever $\rho < \frac{\sqrt{5}}{3}$. $\square$

2.6. Proof of Theorem 1.11. Fix $v > 0$ and let $c \in [0, v)$ vary, with $s = s(c) = \sqrt{v^2 - c^2}$, $A = 1 + v$ and $\Phi(c) = \log \mathcal{G}_*(x, y)/c^2$ for the pair $x = v + c$, $y = v - c$. Put

$$R(c) := \zeta(2, A - \Delta(c)) - \Phi(c), \quad \Delta(c) = \frac{c^2}{3(v+s)} = \frac{v-s}{3}.$$

Since $z \mapsto \zeta(2, z)$ is strictly decreasing and $\zeta(2, 1+t) = \Phi(c)$, while $\frac{2v+s}{3} = A - \Delta$, we have

$$(37) \quad \theta > \frac{2}{3} \iff t > \frac{2v+s}{3} \iff \zeta(2, A - \Delta) > \Phi \iff R(c) > 0.$$

By Theorem 1.1, $\Phi(c) = \sum_{k \geq 1} \frac{c^{2k-2}}{k} \zeta(2k, A)$ is a power series in c^2 whose radius of convergence is $A > v$; hence R is differentiable on $(0, v)$ and $R(0) = 0$. It therefore suffices to prove $R'(c) > 0$ for $0 < c < v$. Using $\Delta' = \frac{c}{3s}$ and $\partial_\Delta \zeta(2, A - \Delta) = 2\zeta(3, A - \Delta)$,

$$R'(c) = 2\zeta(3, A - \Delta(c)) \Delta'(c) - \Phi'(c) = \frac{2c\zeta(3, A - \Delta(c))}{3s} - \frac{2}{c}E(c),$$

where $E(c) := \sum_{m \geq 1} \frac{m}{m+1} c^{2m} \zeta(2m+2, A)$. Since $n + A \geq A$, we have $\zeta(2m+2, A) \leq A^{-(2m-2)} \zeta(4, A)$ for $m \geq 1$, so with $\rho = c/A$,

$$E(c) \leq c^2 \zeta_4 \sum_{m \geq 1} \frac{m}{m+1} \rho^{2m-2} \leq c^2 \zeta_4 \left(\frac{1}{2} + \frac{\rho^2}{1-\rho^2} \right) = \frac{c^2 \zeta_4 (1+\rho^2)}{2(1-\rho^2)}.$$

Moreover $\zeta(3, A - \Delta) > \zeta(3, A)$, and the trapezoid and midpoint rules applied to the convex decreasing functions $t \mapsto (t+A)^{-3}$ and $t \mapsto (t+A)^{-4}$ give

$$\zeta(3, A) \geq \int_0^\infty \frac{dt}{(t+A)^3} + \frac{1}{2A^3} = \frac{A+1}{2A^3}, \quad \zeta(4, A) \leq \int_{-1/2}^\infty \frac{dt}{(t+A)^4} = \frac{1}{3(A - \frac{1}{2})^3},$$

whence

$$(38) \quad \frac{\zeta(4, A)}{\zeta(3, A)} \leq \frac{2A^3}{3(A - \frac{1}{2})^3(A+1)}.$$

Inserting these two estimates, $R'(c) > 0$ holds as soon as $\frac{1}{s} > \frac{A^3(1+\rho^2)}{(A - \frac{1}{2})^3(A+1)(1-\rho^2)}$, and since $s \leq v = A - 1$ it is enough that $\Psi(A) \frac{1+\rho^2}{1-\rho^2} < 1$, where $\Psi(A) = \frac{(A-1)A^3}{(A - \frac{1}{2})^3(A+1)}$. The elementary inequality

$$(39) \quad \Psi(A) \leq 1 - \frac{1}{3A} \quad (A > 1)$$

holds: the difference between the two sides equals $\frac{4A^4 - 14A^3 + 21A^2 - 8A + 1}{3A(A+1)(2A-1)^3}$, and substituting $t = A - 1 > 0$ its numerator becomes $4t^4 + 2t^3 + 3t^2 + 8t + 4 > 0$, all coefficients being positive. Consequently $R'(c) > 0$ whenever $\rho^2 < \frac{1}{6A-1}$, that is for all c with $c^2 < \frac{A^2}{6A-1}$; as $\frac{A^2}{6A-1} > \frac{A}{6}$, this covers every c with $c^2 \leq \frac{A}{6}$, which is (31). Hence $R > 0$ on $(0, c]$ and, by (37), $\theta > \frac{2}{3}$. $\square$

2.7. Proof of Theorem 1.6. If $c_j = 0$, i.e. $x_j = y_j$ for some j , then $\log \mathcal{G}_*(x_j, y_j) = 0$ and the assertion is trivial, so we may assume $c_j > 0$. Put $A_j = 1 + v_j$, $c_j = |u_j|$, $\rho_j = c_j/A_j \in (0, 1)$, so $\rho_j \rightarrow \rho_0$ and $\Phi_j := \frac{\log \mathcal{G}_*(x_j, y_j)}{c_j^2} = \sum_{k \geq 1} \frac{c_j^{2k-2}}{k} \zeta(2k, A_j)$. Set $h_k(A) := \frac{A^{2k-1}}{k} \zeta(2k, A)$, so that $A_j \Phi_j = \sum_{k \geq 1} \rho_j^{2k-2} h_k(A_j)$. Since $\frac{A^{2k-1}}{(n+A)^{2k}} = \frac{1}{A} \left(\frac{A}{n+A} \right)^{2k}$ and $t \mapsto \left(\frac{A}{A+t} \right)^{2k}$ is decreasing, the integral test gives, for all $k \geq 1$ and $A > 0$,

$$(40) \quad 0 \leq h_k(A) = \frac{1}{kA} \sum_{n \geq 0} \left(\frac{A}{n+A} \right)^{2k} \leq \frac{1}{kA} \left(1 + \int_0^\infty \left(\frac{A}{A+t} \right)^{2k} dt \right) = \frac{1}{kA} + \frac{1}{k(2k-1)}.$$

Hence, for any ρ_1 with $\rho_0 < \rho_1 < 1$ and all large j ,

$$\rho_j^{2k-2} h_k(A_j) \leq \rho_1^{2k-2} \left(\frac{1}{k} + \frac{1}{k(2k-1)} \right) \quad (k \geq 1),$$

and the dominating series converges, so dominated convergence applies. On the other hand, for each fixed k the Riemann sum $\frac{1}{A} \sum_{n \geq 0} \left(\frac{A}{n+A}\right)^{2k}$ converges to $\int_0^\infty (1+t)^{-2k} dt = \frac{1}{2k-1}$ as $A \rightarrow \infty$, so $h_k(A) \rightarrow \frac{1}{k(2k-1)}$. Therefore, by the definition (17) of H ,

$$A_j \Phi_j \longrightarrow \sum_{k \geq 1} \frac{\rho_0^{2k-2}}{k(2k-1)} = H(\rho_0),$$

the case $\rho_0 = 0$ being the definition $H(0) = 1$, and the case $\rho_0 > 0$ following from the identity $\sum_{k \geq 1} z^{k-1}/(k(2k-1)) = \Lambda(\sqrt{z})/z$ for $0 < z < 1$, which is obtained by integrating $\sum_{k \geq 1} z^{k-1}/(2k-1) = \operatorname{artanh} \sqrt{z}/\sqrt{z}$ term by term. Since $c_j = A_j \rho_j$, the identity $\log \mathcal{G}_\star = c_j^2 \Phi_j$ reads $\log \mathcal{G}_\star = \rho_j^2 A_j \cdot A_j \Phi_j$; as H is continuous at ρ_0 ,

$$\log \mathcal{G}_\star(x_j, y_j) = \frac{u_j^2}{1+v_j} H(\rho_j)(1+o(1)) = (1+v_j) \Lambda(\rho_j)(1+o(1)),$$

the second equality being $\Lambda(\rho) = \rho^2 H(\rho)$. The proof for $\mathcal{G}$ is the same with A_j replaced by v_j and ρ_j by $|u_j|/v_j$, and gives $\log \mathcal{G} = \frac{u_j^2}{v_j} H\left(\frac{|u_j|}{v_j}\right)(1+o(1))$. $\square$

2.8. Proof of Theorem 1.7. Each x_i satisfies $|x_i - \bar{x}| < \bar{x}$, so x_i lies in the disc $D(\bar{x}, \bar{x}) \subset \{\operatorname{Re} z > 0\}$ on which $\log \Gamma$ is holomorphic with radius of convergence exactly $\bar{x}$. Hence

$$\log \Gamma(x_i) = \log \Gamma(\bar{x}) + \sum_{m \geq 1} \frac{\psi^{(m-1)}(\bar{x})}{m!} (x_i - \bar{x})^m$$

converges for every i . Summing over i and using $\sum_i (x_i - \bar{x}) = 0$,

$$\sum_{i=1}^n \log \Gamma(x_i) - n \log \Gamma(\bar{x}) = \sum_{m \geq 2} \frac{\psi^{(m-1)}(\bar{x})}{m!} m_m = \sum_{m \geq 2} \frac{(-1)^m}{m} \zeta(m, \bar{x}) m_m,$$

which is (18); absolute convergence follows from $|m_m| \leq n \max_i |x_i - \bar{x}|^m$ and $|x_i - \bar{x}| < \bar{x}$. The inequality is Jensen's inequality for the convex function $\log \Gamma$, with equality only when all x_i coincide. $\square$

2.9. Proof of Theorem 1.8. (i) *Uniqueness.* The function $g(z) = \zeta(2, 1+z)$ is continuous and strictly decreasing on $(0, \infty)$. By (8) and the positivity of the terms of (13),

$$\left(\frac{x-y}{2}\right)^2 \zeta(2, 1+v) \leq \log \mathcal{G}_\star(x, y) \leq \left(\frac{x-y}{2}\right)^2 \zeta(2, 1+s),$$

so $\Phi := \log \mathcal{G}_\star(x, y)/c^2$ satisfies $g(v) \leq \Phi \leq g(s)$, both inequalities being strict for $x \neq y$. Hence exactly one $t \in (s, v)$ satisfies $g(t) = \Phi$.

(ii) *The bracket.* Write $\delta = v - t \in (0, v - s)$ and $A = 1 + v$. From (13),

$$(41) \quad \Phi = \zeta(2, A) + \frac{c^2}{2} \zeta(4, A) + T', \quad T' := \sum_{k \geq 3} \frac{c^{2k-2}}{k} \zeta(2k, A) \geq 0.$$

Taylor expansion of $z \mapsto \zeta(2, z)$ about A , all of whose terms are positive, gives

$$(42) \quad \zeta(2, A - \delta) = \zeta(2, A) + 2\delta \zeta(3, A) + 3\delta^2 \zeta(4, A) + S', \quad S' := \sum_{j \geq 3} (j+1) \delta^j \zeta(j+2, A) \geq 0.$$

Since $\zeta(2, A - \delta) = \Phi$, subtraction yields

$$(43) \quad 2\delta \zeta(3, A) + 3\delta^2 \zeta(4, A) + S' = \frac{c^2}{2} \zeta(4, A) + T'.$$

For the lower bound write $\zeta_j := \zeta(j, A)$ and put

$$(44) \quad \delta_0 := \frac{c^2 \zeta_4}{4\zeta_3 + 6v\zeta_4}.$$

We prove $\Phi > \zeta(2, A - \delta_0)$, which by strict decrease of $z \mapsto \zeta(2, z)$ gives $\delta > \delta_0$, i.e. the lower bound in (19). By (41) we have $\Phi \geq \zeta_2 + \frac{c^2}{2}\zeta_4$ (as $T' \geq 0$), while by (42), $\zeta(2, A - \delta_0) = \zeta_2 + 2\delta_0\zeta_3 + 3\delta_0^2\zeta_4 + S'_0$ with $S'_0 := \sum_{j \geq 3} (j+1)\delta_0^j\zeta_{j+2} \geq 0$. It therefore suffices to verify

$$(45) \quad S'_0 < \frac{c^2}{2}\zeta_4 - 2\delta_0\zeta_3 - 3\delta_0^2\zeta_4 = 3\delta_0\zeta_4(v - \delta_0),$$

the equality following from (44) together with $\delta_0(2\zeta_3 + 3v\zeta_4) = \frac{c^2}{2}\zeta_4$.

Now $\delta_0 < \frac{c^2\zeta_4}{6v\zeta_4} = \frac{c^2}{6v} < \frac{v}{6}$, so $\delta_0 < \frac{v}{6} < v < A$. Put $\sigma := \delta_0/A \in (0, \frac{1}{6})$. Since $n + A \geq A$ for $n \geq 0$, we have $\zeta_{j+2} \leq A^{-(j-2)}\zeta_4$ for $j \geq 2$, and hence

$$S'_0 \leq \zeta_4 A^2 \sum_{j \geq 3} (j+1)\sigma^j = \zeta_4 A^2 \frac{\sigma^3(4-3\sigma)}{(1-\sigma)^2}.$$

As $\sigma < \frac{1}{6}$ we have $\frac{4-3\sigma}{(1-\sigma)^2} \leq \frac{4}{(5/6)^2} = \frac{144}{25}$, and as $\delta_0 < \frac{v}{6}$ we have $3(v - \delta_0) > \frac{5v}{2}$. Moreover $\delta_0 < \frac{c^2}{6v} < \frac{v}{6}$, so $\delta_0^2 < \frac{v^2}{36}$. Therefore (45) follows from

$$\zeta_4 \frac{\delta_0^2}{A} \cdot \frac{144}{25} \leq \zeta_4 \cdot \frac{5v}{2}, \quad \text{which is implied by} \quad \frac{144}{25} \cdot \frac{v^2}{36A} \leq \frac{5v}{2},$$

and the last inequality reads $\frac{4v^2}{25A} \leq \frac{5v}{2}$, i.e. $\frac{8v}{25} \leq 5(1+v)$, which holds for every $v > 0$. This proves the lower bound.

For the upper bound we estimate T' . Since $n + A \geq A$, for $k \geq 2$ and $n \geq 0$ we have $(n + A)^{-2k} \leq A^{-(2k-4)}(n + A)^{-4}$, hence $\zeta(2k, A) \leq A^{-(2k-4)}\zeta(4, A)$; therefore, substituting $k = j + 3$ in the sum,

$$T' \leq \zeta(4, A) \sum_{k \geq 3} \frac{c^{2k-2}}{kA^{2k-4}} = \frac{\zeta(4, A)c^4}{A^2} \sum_{j \geq 0} \frac{\rho^{2j}}{j+3} \leq \frac{\zeta(4, A)c^4}{A^2} \cdot \frac{1}{3(1-\rho^2)} = \frac{c^2}{2}\zeta(4, A) \cdot \frac{2\rho^2}{3(1-\rho^2)},$$

where we used $\frac{1}{j+3} \leq \frac{1}{3}$ and $\rho < 1$. Inserting this into (43) and dropping $S' \geq 0$ gives $2\delta\zeta(3, A) \leq \frac{c^2}{2}\zeta(4, A)(1 + \frac{2\rho^2}{3(1-\rho^2)})$, the upper bound in (19).

(iii) *Asymptotics.* Write $\zeta_j := \zeta(j, A)$. From $\Phi = \zeta_2 + \frac{c^2}{2}\zeta_4 + \frac{c^4}{3}\zeta_6 + O(c^6)$ and $\zeta(2, A - \delta) = \zeta_2 + 2\delta\zeta_3 + 3\delta^2\zeta_4 + O(\delta^3)$, equating and inserting $\delta = \alpha c^2 + \beta c^4 + O(c^6)$ gives $2\zeta_3\alpha = \frac{\zeta_4}{2}$ and $2\zeta_3\beta + 3\zeta_4\alpha^2 = \frac{\zeta_6}{3}$, that is $\alpha = \frac{\zeta_4}{4\zeta_3}$ and $\beta = \frac{1}{2\zeta_3}[\frac{\zeta_6}{3} - \frac{3\zeta_4^3}{16\zeta_3^3}]$, which is (20).

(iv) *Condition (A).* By (ii) it suffices that $\frac{c^2\zeta_4}{4\zeta_3}(1 + \frac{2\rho^2}{3(1-\rho^2)}) < \frac{v-s}{2} = \frac{c^2}{2(v+s)}$, i.e. $\frac{(v+s)\zeta_4}{2\zeta_3}(1 + \frac{2\rho^2}{3(1-\rho^2)}) < 1$. Term-wise, $(v+s)\zeta_4 \leq \frac{v+s}{1+v}\zeta_3$, because $\frac{v+s}{(n+A)^4} \leq \frac{2}{(n+A)^3}$ is equivalent to $v+s \leq 2(n+1+v)$, true for all $n \geq 0$ since $s < v$. Hence (21) is sufficient.

(iv) *Condition (B).* Put $K(\xi) = \frac{1}{c^2} \log \frac{1}{1-c^2/\xi^2}$ and $L(\xi) = (\xi - w)^{-2}$ with $w = \frac{v-s}{2}$. With $a = A = 1 + v$,

$$\sum_{m \geq 1} K(m+v) = \sum_{n \geq 0} K(a+n) = \Phi, \quad \sum_{m \geq 1} L(m+v) = \sum_{n \geq 0} \frac{1}{(n+1 + \frac{v+s}{2})^2} = \zeta\left(2, 1 + \frac{v+s}{2}\right),$$

so $t > \frac{v+s}{2} \iff \Phi < \zeta\left(2, 1 + \frac{v+s}{2}\right) \iff \sum_{n \geq 0} (K - L)(a+n) < 0$. On $[v + \frac{1}{2}, \infty)$ both K and L are positive, decreasing and convex: $L'' = 6(\xi - w)^{-4} > 0$, while $K'(\xi) = -\frac{2}{\xi^3 - c^2\xi}$ and $K''(\xi) = \frac{2(3\xi^2 - c^2)}{(\xi^3 - c^2\xi)^2} > 0$ because $\xi \geq v + \frac{1}{2} > v > c > c/\sqrt{3}$. For a convex decreasing f with $f(\infty) = 0$ the midpoint rule $\int_{k-1/2}^{k+1/2} f \geq f(k)$ gives $\sum_{n \geq 0} f(a+n) \leq \int_{a-1/2}^{\infty} f$, and the trapezoid rule $\int_k^{k+1} f \leq \frac{f(k)+f(k+1)}{2}$ gives $\sum_{n \geq 0} f(a+n) \geq \int_a^{\infty} f + \frac{f(a)}{2}$. Applying these to K and L ,

$$\sum_{n \geq 0} (K - L)(a+n) \leq \int_{v+1/2}^{\infty} K - \int_a^{\infty} L - \frac{L(a)}{2} = \int_{v+1/2}^{\infty} K - \frac{1}{A-w} - \frac{1}{2(A-w)^2} = \mathcal{E}(x, y),$$

and $\int_B^\infty K$ is (23), obtained by elementary integration. Hence $\mathcal{E}(x, y) < 0$ is sufficient for $t > \frac{v+s}{2}$. $\square$

3. APPLICATIONS

3.1. Sharpened Gurland inequalities. Since $\log \mathcal{G}(x, y) = \sum_{k \geq 1} \frac{u^{2k}}{k} \zeta(2k, v)$ with all terms positive, and since $\zeta(2k, v) \leq v^{-2k} (1 + \frac{v}{2k-1})$ by the integral test, we obtain a two-sided refinement of Gurland's inequality [5].

Corollary 3.1. *For all $x \neq y > 0$, with $v = \frac{x+y}{2}$ and $\rho_G = \frac{|x-y|}{x+y}$,*

$$\frac{(x-y)^2}{4} \zeta\left(2, \frac{x+y}{2}\right) \leq \log \mathcal{G}(x, y) \leq -\log(1 - \rho_G^2) + v \Lambda(\rho_G),$$

and $\mathcal{G}(x, y) > \exp\left(\frac{(x-y)^2}{4} \zeta\left(2, \frac{x+y}{2}\right)\right) > 1$. Moreover $\log \mathcal{G}(x, y) > \frac{(x-y)^2}{2(x+y)}$.

Proof. The lower bound is the first term of (12). For the upper bound, the terms of $\zeta(2k, v) = \sum_{n \geq 0} (n+v)^{-2k}$ with $n \geq 1$ are dominated by the integral over $[0, \infty)$: since $(t+v)^{-2k}$ decreases, $\int_{n-1}^n (t+v)^{-2k} dt > (n+v)^{-2k}$ for every $n \geq 1$, so $\zeta(2k, v) \leq v^{-2k} + \int_0^\infty (t+v)^{-2k} dt = v^{-2k} (1 + \frac{v}{2k-1})$; then, with $\rho = \rho_G$,

$$\log \mathcal{G}(x, y) \leq \sum_{k \geq 1} \frac{u^{2k}}{k} \left(\frac{1}{v^{2k}} + \frac{v^{1-2k}}{2k-1} \right) = \sum_{k \geq 1} \frac{\rho^{2k}}{k} + v \sum_{k \geq 1} \frac{\rho^{2k}}{k(2k-1)} = -\log(1 - \rho^2) + v \Lambda(\rho).$$

The last claim uses $\zeta(2, v) = \sum_{n \geq 0} (n+v)^{-2} > \int_0^\infty (t+v)^{-2} dt = \frac{1}{v}$, because $\int_n^{n+1} (t+v)^{-2} dt < (n+v)^{-2}$ for every $n \geq 0$. $\square$

3.2. Bounds for the beta function. Recall $B(x, y) = \Gamma(x)\Gamma(y)/\Gamma(x+y)$ and the duplication formula $\Gamma(2v) = \frac{2^{2v-1}}{\sqrt{\pi}} \Gamma(v)\Gamma(v + \frac{1}{2})$, where $v = \frac{x+y}{2}$. Hence $B(x, y) = \mathcal{G}(x, y) \frac{\sqrt{\pi} \Gamma(v)}{2^{2v-1} \Gamma(v + \frac{1}{2})}$.

Combining Corollary 3.1 with Gautschi's inequality $v^{1/2} < \frac{\Gamma(v+1)}{\Gamma(v+1/2)} < (v+1)^{1/2}$ [4] yields the following.

Corollary 3.2. *For all $x, y > 0$, with $v = \frac{x+y}{2}$ and $\rho_G = \frac{|x-y|}{x+y}$,*

$$\frac{\sqrt{\pi} v^{-1/2}}{2^{2v-1}} e^{\frac{(x-y)^2}{4} \zeta(2, v)} < B(x, y) < \frac{\sqrt{\pi} (v+1)^{1/2}}{2^{2v-1} v} e^{v \Lambda(\rho_G) - \log(1 - \rho_G^2)}.$$

Proof. Gautschi's inequality with $s = \frac{1}{2}$ and $x = v$ reads $v^{1/2} < \frac{\Gamma(v+1)}{\Gamma(v+1/2)} < (v+1)^{1/2}$; since $\Gamma(v+1) = v\Gamma(v)$, this gives $v^{-1/2} < \frac{\Gamma(v)}{\Gamma(v+1/2)} < \frac{(v+1)^{1/2}}{v}$. Insert this into the expression for $B(x, y)$ and use Corollary 3.1. $\square$

Example 3.3. For $(x, y) = (1, 2)$ one has $v = \frac{3}{2}$, $\rho_G = \frac{1}{3}$, $\mathcal{G}(1, 2) = 1.273240\dots$ and $B(1, 2) = \frac{1}{2}$. The bounds of Corollary 3.2 read $0.457050 < 0.5 < 0.622777$.

3.3. Certified computation. The two-sided estimate (15) turns the expansion into an algorithm with a certified error.

Corollary 3.4. *Let $x \neq y > 0$, $\rho = \rho(x, y) \in (0, 1)$, $A = 1 + v$ and $0 < \varepsilon < \frac{1}{2}$. If*

$$m \geq \left\lceil \frac{\log((1+A)/(\varepsilon(1-\rho^2)))}{2 \log(1/\rho)} \right\rceil,$$

then $|\log \mathcal{G}_*(x, y) - S_m(x, y)| \leq \varepsilon$. Moreover, for fixed x, y the number of terms required does not exceed

$$\frac{\log(1/\varepsilon) + \log(2+v) + \frac{1}{1-\rho}}{2(1-\rho)},$$

so it is $O(\frac{\log(1/\varepsilon)}{1-\rho})$ as $\varepsilon \rightarrow 0$, the implied constant being absolute; the arguments enter only through ρ and through the additive term $\log(2+v)$, which is independent of ε .

Proof. For every $m \geq 1$ we have $2m - 1 \geq 1$, hence $1 + \frac{A}{2m-1} \leq 1 + A$ and $\frac{1}{m} \leq 1$; therefore (15) gives $0 \leq R_m \leq \frac{(1+A)\rho^{2m}}{1-\rho^2}$, which is at most ε whenever $\rho^{2m} \leq \frac{\varepsilon(1-\rho^2)}{1+A}$. Taking logarithms gives the stated threshold. For the estimate, use $\log \frac{1}{1-\rho^2} \leq \frac{1}{1-\rho}$ (valid for $0 \leq \rho < 1$, since $\log \frac{1}{1-\rho^2} \leq \frac{\rho^2}{1-\rho^2} \leq \frac{1}{1-\rho}$) and $-\log \rho \geq 1 - \rho$; then

$$m \leq \frac{\log(1+A) + \log \frac{1}{\varepsilon} + \log \frac{1}{1-\rho^2}}{2 \log \frac{1}{\rho}} \leq \frac{\log(1/\varepsilon) + \log(2+v) + \frac{1}{1-\rho}}{2(1-\rho)}.$$

The leading term as $\varepsilon \rightarrow 0$ is $\frac{\log(1/\varepsilon)}{2(1-\rho)}$, with the absolute constant $\frac{1}{2}$. $\square$

3.4. The multivariate Jensen–Gurland inequality. Theorem 1.7 with $n = 2$ reproduces Theorem 1.1. For a concrete three-point case take $0 < \varepsilon < \frac{1}{2}$ and

$$x_1 = x_2 = v(1 + \varepsilon), \quad x_3 = v(1 - 2\varepsilon),$$

so that $\bar{x} = v$ and $m_j = v^j \varepsilon^j (2 + (-2)^j)$; (18) becomes

$$\log \frac{\Gamma(v(1 + \varepsilon))^2 \Gamma(v(1 - 2\varepsilon))}{\Gamma(v)^3} = 3v^2 \varepsilon^2 \zeta(2, v) + 2v^3 \varepsilon^3 \zeta(3, v) + \frac{9}{2} v^4 \varepsilon^4 \zeta(4, v) + O(\varepsilon^5),$$

all displayed coefficients being positive, in accordance with the multivariate Jensen–Gurland inequality of Theorem 1.7.

4. NUMERICAL VERIFICATION

All computations were performed with `mpmath` at 35–60 decimal digits by a single self-contained script `verify.py`, whose nine sections correspond to the numbered statements below and which reports `ok`/`FAIL` for each of the 50 checks (all pass). Table 1 compares the exact expansion (13) with $\log \mathcal{G}_\star$ computed from $\log \Gamma$; Table 2 checks the two-sided estimate (15), including the extreme case $(0.1, 199.9)$ where $\rho = 0.9891$; Table 3 checks the bracket (19); Table 4 documents that the two sufficient conditions of Theorem 1.8(iv) cover the whole sampled parameter range; Table 5 traces the approach of the ratio (24) to its sharp value $\frac{2}{3}$, which is reached only in the joint limit $v \rightarrow \infty$, $\varepsilon \rightarrow 0$, $v\varepsilon \rightarrow 0$; and Table 6 records an independent Euler–Maclaurin certificate for the midpoint property.

TABLE 1. Exact expansion: $|\log \mathcal{G}_\star - S_m|$ for the truncated series (13), and the number m of terms used.

x	y	$ \log \mathcal{G}_\star - S_m $	m
1	2	$3.5 \cdot 10^{-51}$	34
1	10	$5.1 \cdot 10^{-48}$	142
0.5	3	$1.0 \cdot 10^{-49}$	68
0.2	0.9	$2.1 \cdot 10^{-51}$	37
3	3.0001	$1.0 \cdot 10^{-61}$	6

TABLE 2. The two-sided estimate (15): $\text{lo} = \frac{u^{2m}}{m} \zeta(2m, 1 + v)$ and hi is the right-hand side of (15).

(x, y)	ρ	m	R_m	lo	hi
(1, 10)	0.6923	2	$4.1203 \cdot 10^{-1}$	$3.1211 \cdot 10^{-1}$	$6.9851 \cdot 10^{-1}$
(1, 10)	0.6923	12	$2.2659 \cdot 10^{-5}$	$1.2662 \cdot 10^{-5}$	$3.0165 \cdot 10^{-5}$
(0.5, 3)	0.4545	2	$3.6843 \cdot 10^{-2}$	$3.2677 \cdot 10^{-2}$	$5.1563 \cdot 10^{-2}$
(0.5, 3)	0.4545	12	$6.2365 \cdot 10^{-10}$	$5.0458 \cdot 10^{-10}$	$7.1160 \cdot 10^{-10}$
(0.1, 199.9)	0.9891	12	2.9021	0.3147	15.9456

TABLE 3. The bracket (19) of Theorem 1.8: $\delta = v - t$ against its lower bound lo and the two-term value pre = $\frac{c^2\zeta(4,A)}{4\zeta(3,A)}$; the upper bound of (19) is never violated.

(x, y)	ρ	$\delta = v - t$	lo	pre
(1, 2)	0.2000	0.0199947	0.0115425	0.0197486
(0.5, 3)	0.4545	0.1186430	0.0635181	0.1108211
(2, 7)	0.4545	0.2194361	0.1090893	0.2063217
(1, 10)	0.6923	0.6573045	0.2924374	0.5586888
(20, 100)	0.6557	4.9818888	2.2129255	4.4074125
(0.1, 199.9)	0.9891	26.6699099	8.2958322	16.5501874
(100, 101)	0.0049	0.0004125	0.0002068	0.0004125

TABLE 4. Coverage of Theorem 1.8(iv). The first family is the coarse 11-point grid $\{10^{-3}, 10^{-2}, 10^{-1}, \frac{1}{2}, 1, 2, 5, 20, 10^2, 10^3, 10^4\}^2$, whose 121 points consist of 11 diagonal pairs $x = y$ and 110 off-diagonal pairs. On the diagonal $\rho = c/A = 0$ and both halves of (19) degenerate, so only the 110 off-diagonal pairs are counted, and that is the number reported in the cases column. The other two families are $x = v(1 + r)$, $y = v(1 - r)$ with $r = j/400$, $j = 1, \dots, 399$. For those, $\rho(x, y) = \frac{vr}{1+v}$ runs over $(0, \frac{v}{1+v})$ in steps of $\frac{v}{400(1+v)} < \frac{1}{400}$, so the range of ρ depends on v . A configuration may satisfy both conditions, so the (A) and (B) columns need not add up to the number of cases; in the coarse grid 22 of the 110 off-diagonal pairs satisfy both, and in every sample at least one of the two conditions holds, so the midpoint property is certified throughout.

sample	cases	(A)	(B)	neither
coarse grid	110	42	90	0
$v = 200$	399	43	396	0
$v = 5000$	399	8	399	0

TABLE 5. The ratio $\theta = (t - \sqrt{xy}) / (\frac{x+y}{2} - \sqrt{xy})$ along $x = v(1 + \varepsilon)$, $y = v(1 - \varepsilon)$. For a fixed ε it decreases to $\frac{2}{3} + g(\varepsilon)$ with $g(\varepsilon) > 0$ (numerically $g(\varepsilon) \approx 5.6\varepsilon^2$, for instance $g(10^{-1}) = 5.60 \cdot 10^{-5}$), so the sharp value $\frac{2}{3}$ is approached only in the joint limit $v \rightarrow \infty$, $\varepsilon \rightarrow 0$, $v\varepsilon \rightarrow 0$. The penultimate column is such a path ($\varepsilon = v^{-2}$), and there $\theta = \frac{2}{3} + \frac{1}{6v} + O(v^{-2})$, as the elementary computation in the remark after Theorem 1.9 shows.

v	$\varepsilon = 10^{-1}$	$\varepsilon = 10^{-3}$	$\varepsilon = v^{-2}$	limit
10^2	0.668396737	0.668333257	0.668333250	0.666666667
10^3	0.666890050	0.666833339	0.666833333	0.666666667
10^4	0.666739371	0.666683339	0.666683333	0.666666667
10^5	0.666724303	0.666668339	0.666668333	0.666666667
10^6	0.666722796	0.666666839	0.666666833	0.666666667
10^7	0.666722645	0.666666689	0.666666683	0.666666667

Several further checks are worth recording. First, for $(x, y) = (100, 1)$ one has $Q(100, 1) = 4.5 > 1$, so Corollary 1 of [9] does not apply, yet $\rho(100, 1) = 0.96117 < 1$ and the expansion (13) converges, with tails $R_{20} = 1.478 \cdot 10^{-1}$, $R_{100} = 4.359 \cdot 10^{-5}$ and $R_{1000} = 5.117 \cdot 10^{-37}$. Second, the optimality statement in Theorem 1.3 is confirmed: for $(x, y) = (2, 7)$ one has $2 \log \rho = -1.5769147$, while the computed values of $\frac{1}{m} \log R_m$ are -1.8184 ($m = 5$), -1.7829 ($m = 10$), -1.7157 ($m = 20$), -1.6635 ($m = 40$), -1.6288 ($m = 80$) and -1.6072 ($m = 160$), approaching the predicted limit monotonically. Third, the three-term expansion (20) is

TABLE 6. Euler–Maclaurin certificate for the midpoint property, evaluated on the 110 off-diagonal pairs of the coarse grid of Table 4 (first three rows) and on four further pairs (last row). With $v = \frac{x+y}{2}$, $c = \frac{|x-y|}{2}$, $s = \sqrt{xy}$, $a = A = 1 + v$, write $G = K - L$, where $K(\xi) = \frac{1}{c^2} \log \frac{1}{1-c^2/\xi^2}$ and $L(\xi) = (\xi - w)^{-2}$ with $w = \frac{v-s}{2}$. The partial sum $E_r = \int_a^\infty G + \frac{G(a)}{2} - \sum_{j=1}^r \frac{B_{2j}}{(2j)!} G^{(2j-1)}(a)$ approximates $\Sigma = \sum_{n \geq 0} G(a+n)$, and the remainder obeys the classical bound $|R_r| \leq \frac{2\zeta(2r)}{(2\pi)^{2r}} \int_a^\infty |G^{(2r)}|$, because the periodic Bernoulli polynomial $\tilde{B}_{2r}$ attains its extreme value B_{2r} at the endpoints. Since $\Sigma < 0$ is exactly the midpoint property (of which condition (B) of Theorem 1.8 is the sufficient criterion $\Sigma \leq \mathcal{E} < 0$), the table certifies the midpoint property for each sampled configuration directly, without passing through the midpoint and trapezoid estimates used there. The last row compares E_r with Σ computed independently, as the difference between the exact value of $\sum_{m \geq 1} L(m+v) = \zeta(2, 1 + \frac{v+s}{2})$ and a cancellation-free Euler–Maclaurin evaluation of $\sum_{m \geq 1} K(m+v)$; it is evaluated for all six orders $r = 1, \dots, 6$, although only $r \leq 5$ certify negativity for every sampled configuration.

quantity	value	attained at
worst $E_r + R_r $, $r = 1, \dots, 5$	$-1.9844 \cdot 10^{-5}$	(1000, 10000)
configurations with $E_r + R_r \geq 0$	0 of 110	—
largest $ E_r - \Sigma / R_r $, $r = 1, \dots, 6$	0.72	(0.01, 10^4)

confirmed: for $(x, y) = (10, 11)$ one has $v - t = 0.00378192138106$, while the two-term formula gives 0.00378011106367 and the three-term formula 0.00378191965719, the residual being $-1.7 \cdot 10^{-9}$. Fourth, the large-argument asymptotics of Theorem 1.6: for $(x, y) = (10^6, 2 \cdot 10^6)$, where $v = 1.5 \cdot 10^6$ and $\rho = \frac{|x-y|}{2+x+y} = 0.333333\dots$, one has $\log \mathcal{G}_\star = 169898.977904\dots$ against the prediction $(1+v)\Lambda(\rho) = \frac{u^2}{1+v}H(\rho) = 169898.919012\dots$, a relative error of $3.5 \cdot 10^{-7}$, which decreases as the arguments grow ($2.7 \cdot 10^{-5}$ at $(10^4, 3 \cdot 10^4)$). Fifth, Corollaries 3.1 and 3.2 were verified on the sampled configurations; for instance $B(1, 2) = \frac{1}{2}$ lies between 0.457050 and 0.622777, and the three-point expansion displayed in Section 3 agrees with $\log \Gamma$ to 19 digits.

5. CONCLUDING REMARKS

Theorem 1.1 is the classical Hurwitz-zeta expansion (11) of $\log \Gamma$, specialized to the second-order symmetric difference that defines $\log \mathcal{G}_\star$. In that specialization it identifies the mechanism behind the expansion of [9]: the logarithm of Gurland's ratio is a Taylor remainder about the midpoint, and the poles of Γ at the non-positive integers determine both the coefficients (Hurwitz zeta values) and the exact region of convergence, which is the entire quadrant. Two of the three problems posed in [9] are thereby answered affirmatively, and the quantitative content of the third is the subject of Theorems 1.8 and 1.9. For the third we proved uniqueness of t , its expansion up to order c^4 , an explicit two-sided bracket, the midpoint property $t > \frac{1}{2}(\sqrt{xy} + \frac{x+y}{2})$ under either of the two complementary conditions (A) and (B) of Theorem 1.8, and finally the sharp bound $\theta > \frac{2}{3}$ of Theorem 1.9, whose proof uses neither (A) nor (B) and shows the constant $\frac{2}{3}$ to be optimal and not attained. Three directions seem worth pursuing:

- (a) a proof of Conjecture 1.10, i.e. that conditions (A) and (B) cover the whole quadrant; the sharp bound $\frac{2}{3}$ itself is now Theorem 1.9, whose proof uses neither (A) nor (B); the extremal configuration for the sharp constant is $v \rightarrow \infty$ with $|x - y|/(x + y) \rightarrow 0$, as recorded in the remark following Theorem 1.9;
- (b) the q -analogue: replacing Γ by the q -gamma function Γ_q , for which the role of $\zeta(2k, \cdot)$ is played by the Jackson q -zeta values, and the sign-regularity of the corresponding kernel;

- (c) quantitative versions of Theorem 1.7, such as remainder estimates uniform in n and a central limit theorem for the corresponding Jensen gaps.

Statements and declarations. The author has no competing interests to declare. All numerical results reported in Section 4 were produced by the accompanying script `verify.py` and can be reproduced exactly with `mpmath` by running `python verify.py`.

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